

Hotel Booking Demand Clustering Study

A Reproducible Experimental Study of Clustering

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Abstract

This report presents a reproducible clustering study of hotel booking demand data (117,430 records, course release v1) with the goal of partitioning bookings into interpretable guest segments based solely on pre-stay observable features. Two representations are studied: a value-inclusive pipeline (**R-EUCLID-robust-weighted-withADR**) and a purely behavioural pipeline that excludes the daily rate (**R-EUCLID-robust-weighted-noADR**). K-Means ($k = 3$, Silhouette = 0.452, ARI = 0.989 ± 0.007 across 10 seeds), iK-Means (automatic $k = 8$), and Agglomerative Hierarchical clustering (Ward, Silhouette = 0.453 at $k = 3$) are compared under a unified evaluation protocol. The selected three-cluster solution identifies a Non-Refundable Advance Booker segment (12.4%), a Standard Flexible City Guest segment (70.7%), and a Premium Resort Long-Stay segment (16.8%). The noADR comparison confirms that the commitment axis (**is_non_refund**) drives cluster separation independently of price. Key limitations include the absence of ground-truth labels, the use of an ordinal meal-plan encoding, and the fact that clustering is performed at booking level rather than guest level.

Keywords: hotel booking demand; clustering; guest segmentation; mixed-type data; reproducibility; K-Means; Ward linkage

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1 Introduction

Context

Hotel revenue management depends on understanding who books, when, and how. Booking records carry rich pre-arrival signals — rate, stay duration, lead time, deposit commitment, meal plan, parking need, and special requests — that may reveal latent guest types with distinct spending patterns and service preferences. Identifying such types without relying on post-stay outcomes enables targeted pricing, upselling, and capacity planning decisions at the time of booking.

Segmentation question

Can hotel booking records be partitioned into distinct, interpretable behavioural segments using only pre-stay observable characteristics — daily rate, total estimated spend, length of stay, booking horizon, meal plan, deposit commitment, special requests, and parking need — to identify guest profiles that support targeted revenue management and personalised service strategies?

Clustering rationale

No ground-truth segment labels exist; the goal is discovery rather than classification, making supervised methods inapplicable. Booking records exhibit substantial heterogeneity in ADR (€0–5,400), stay duration (0–69 nights), and lead time (0–737 days), suggesting meaningful subgroups. Clustering inputs are restricted to variables known at booking time (leakage control); post-event fields are reserved for post-hoc profiling only.

Contribution

We implement an end-to-end reproducible pipeline covering mixed-type preprocessing, K-Means and Hierarchical Ward clustering, iK-Means initialisation, internal validity assessment (Silhouette, Calinski-Harabasz, Davies-Bouldin), seed-stability analysis (ARI), scaler sensitivity, a PCA representation study (Module E4), and a noADR representation comparison. All results are logged in `experiments.csv` and regenerable via `run.all.py`.

2 Dataset and Preprocessing

2.1 Dataset documentation

Item	Details
Dataset and version	Hotel Booking Demand; course release v1
Unit of analysis	One row = one booking record (a guest may appear multiple times)
Size and time span	119,390 rows, 32 columns; arrivals July 2015 – August 2017
Source and terms	António et al. (2019) [1]; dataset provided through course platform; raw file not committed to repository
Checksum / hash	SHA-256: 7c2ae42a...c7b1fc06
Raw-data handling	Raw CSV excluded via <code>.gitignore</code> ; obtain from course platform and place in <code>data/</code>

2.2 Data quality and feature governance

Missingness and outliers. `children` has 4 missing values (imputed as 0); `country` has 488 missing values (retained for post-hoc profiling only); `agent` (16,340) and `company` (112,593) have high missingness and are dropped as high-cardinality ID-like fields. `adr` has a heavy right tail (max €5,400) and one negative entry; `lead_time` ranges 0–737 days.

Zero-ADR exclusion. Rows with `adr` = 0 (1,960 records, corresponding to complementary or test bookings per dataset documentation) are excluded from clustering, leaving **117,430** records. This exclusion is predefined and reported.

Leakage control. Index time is the booking creation date. The following fields are excluded from clustering inputs:

- `is_canceled`, `reservation_status`, `reservation_status_date` — post-decision / post-arrival outcomes
- `agent`, `company` — high-cardinality ID fields
- `country` — retained for post-hoc profiling only

Feature engineering and encoding.

- `total_nights` = `stays_in_week_nights` + `stays_in_weekend_nights`
- `total_spend_log` = $\log(1 + \text{adr}) \times \text{total_nights}$ — compressed spend proxy combining rate and duration; `adr_log` is *not* included separately to avoid collinearity
- `is_non_refund` = $1[\text{deposit_type} = \text{“Non Refund”}]$ — binary; replaces a three-level ordinal that inflated Euclidean distances
- `lead_time_rank` — `lead_time` binned into three equal-population quantile groups via `pd.qcut(q=3)`: 0 = last-minute (0–32 d), 1 = mid (33–124 d), 2 = advance (≥ 125 d). The equal-population binning is justified over equal-width binning because `lead_time` is heavily right-skewed (0–737 days); quantile bins ensure each rank level represents a comparable share of bookings, making the ordinal distances between levels more meaningful in Euclidean space. A log-transformed continuous alternative was considered but the binned ordinal representation is preferred here to keep the feature on a bounded $[0, 2]$ scale consistent with the other ordinal inputs
- `meal_encoded` — ordinal encoding SC/Undefined= 0, BB= 1, HB= 2, FB= 3, justified by the monotonically increasing catering inclusivity of the four plan levels

Final representations and distance. Two representations are studied:

ID	Features	Scaler
R-EUCLID-robust-weighted-withADR	<code>total_spend_log</code> , <code>total_nights</code> + 5 binary/ordinal	RobustScaler
R-EUCLID-robust-weighted-noADR	<code>total_nights</code> + 5 binary/ordinal	RobustScaler

Both use Euclidean distance. Feature weights `meal_encoded` $\times 0.5$ and `is_non_refund` $\times 3.0$ are applied post-scaling to control relative contribution; this changes the effective Euclidean metric and is recorded as preprocessing. All validity indices are computed in the same weighted space.

3 Clustering Methods

K-Means (baseline). K-Means with `n_init`=10, `max_iter`=300, `random_state`=42. k range 2–10 predefined based on domain plausibility (2 = minimum meaningful split; 10 = upper

limit beyond which segments lose managerial interpretability). Final selection: $k = 3$ based on convergence of Silhouette and ARI stability evidence (see Section 5).

iK-Means. Mirkin’s (2005) intelligent initialisation procedure [2] extracts anomalous clusters greedily from the full 117,430-row matrix with `min_cluster_size=1000` (predefined: clusters below 1,000 bookings lack managerial actionability). The retained k is determined automatically.

Agglomerative Hierarchical Clustering – Ward linkage. Ward linkage minimises within-cluster variance under Euclidean distance, compatible with the RobustScaler + weighted Euclidean representation. Run on a reproducible subsample of $n = 30,000$ (`seed=42`), which represents 25.5% of the analytic dataset. Subsample impact is quantified via stability across five subsamples (`seeds={42, 7, 123, 256, 999}`). k range 2–8; dendrogram cut assessed visually and via validity indices.

4 Experimental Protocol

Experiment block		Purpose	Main output
Baseline (withADR noADR)	K-Means +	Primary segmentation; noADR tests whether clusters are price bands	Validity table, cluster profiles, $\Delta\%$ deviation table
iK-Means		Automatic k estimation; initialisation-free baseline	Anomalous cluster report, validity metrics
Hierarchical Ward		Conceptually distinct family (dendrogram-based)	Dendrogram, validity table, cross-method ARI
Seed stability (K-Means)	(K-Means)	10 seeds $\times k \in \{2, 3, 4, 5\}$	Mean $\pm$ std of Silhouette, CH, DB, and ARI
Subsample stability (Hierarchical)		5 subsamples $\times k = 3$	Mean $\pm$ std of validity indices
Scaler sensitivity		RobustScaler vs StandardScaler on numerical features only	Silhouette delta per k
Unweighted sensitivity		Remove <code>is_non_refund</code> $\times 3.0$ and <code>meal</code> $\times 0.5$; all weights = 1.0	Silhouette and cluster profile comparison vs weighted pipeline
PCA representation study (E4)		Original 7D vs PCA-2, PCA-3, PCA-4	Silhouette comparison, ARI vs original

Validity indices computed in the same weighted Euclidean space for all methods: Silhouette score (primary), Calinski-Harabasz (CH), and Davies-Bouldin (DB). Hierarchical family-specific diagnostic: Ward dendrogram with `truncate_mode=lastp`.

Reproducibility. All seeds are locked (`random_state=42` for main runs; seeds $\{0, \dots, 9\}$ for stability). Results are logged to `experiments.csv` with `representation_id`, `parameters`, `seed`, `silhouette`, `calinski`, `davies`, `diagnostics`, and `notes`. Every figure is regenerable via `run_all.py`.

5 Results and Cluster Interpretation

5.1 Data preparation outcomes

After zero-ADR exclusion: **117,430 bookings**, 7 features (withADR) or 6 features (noADR). The scaled matrix has shape $(117,430 \times 7)$ for withADR and $(117,430 \times 6)$ for noADR. Principal

component analysis of the withADR matrix shows PC1 explains 54.3% and four components are needed to reach 90% variance, confirming moderate linear structure (features are not fully independent).

5.2 Model comparison

Method	Setting	Sil $\uparrow$	CH $\uparrow$	DB $\downarrow$	Stability	
K-Means	$k = 3$, withADR	0.452	63,837	0.857	ARI 0.989 ± 0.007	Best balance
K-Means	$k = 2$, withADR	0.392	54,947	1.074	ARI 0.998 ± 0.002	Merges Premium
K-Means	$k = 3$, noADR	—	—	—	—	Profile
iK-Means	$k = 8$ (auto)	0.327	56,165	1.012	—	Fragments main
Hier. Ward	$k = 3$, subsample	0.453	15,928	0.832	Sil ± 0.004 across 5 subsamples	Corroborates

Selected solution: K-Means $k = 3$ on withADR. Rationale: (i) Silhouette 0.452 exceeds the weak-structure threshold of 0.35 [3]; (ii) ARI 0.989 ± 0.007 confirms near-identical partitions across all 10 seeds; (iii) Hierarchical Ward independently finds the same three- cluster structure (ARI > 0.92 against K-Means $k = 3$); (iv) three segments map to actionable business profiles. iK-Means is not selected as the primary solution because its eight clusters fragment the three main groups without revealing qualitatively new profiles.

5.3 Cluster profiles

Cluster	Size	%	Main characteristics	Label
C2	14,513	12.4	ADR €90, <code>is_non_refund</code> = 1.0, stay 2.7 nights, lead-time rank 1.6 (advance)	Non-Refundable Advance Booker
C1	83,139	70.7	ADR €103, <code>is_non_refund</code> = 0, stay 2.6 nights, lead-time rank 0.8 (last-minute/mid), highest parking need	Standard Flexible City Guest
C0	19,778	16.8	ADR €114, stay 7.7 nights, total spend +136% above mean, most special requests, resort hotel dominant	Premium Resort Long-Stay

Cluster narrative. The Non-Refundable Advance Booker is defined by full deposit commitment (`is_non_refund`= 1.0 for all members), planned booking horizon (lead-time rank +63% above mean), and zero car-parking need — suggesting a city-hotel guest who locks in rates early and forfeits flexibility. Despite having the lowest ADR, this segment represents the highest revenue certainty.

The Standard Flexible City Guest forms the majority segment (70.7%). It has no deposit commitment, a wide range of booking horizons, and short stays typical of business or urban leisure travel. The highest car-parking proportion in this group (+16% above mean) suggests a subset of car-driving guests.

The Premium Resort Long-Stay Guest is the smallest but highest-value segment. Total estimated spend is 136% above the grand mean, driven by both higher ADR and substantially longer stays (7.7 nights vs 2.6 dataset average). Resort hotel is the dominant property type, and above-average special requests (+34%) indicate service-oriented guests who may respond well to upsell offers.

5.4 noADR representation comparison

Removing `total_spend_log` from the feature set and re-clustering tests whether the withADR clusters are purely price bands. Under the noADR representation: the Non-Refundable segment (C2) is preserved intact (high `is_non_refund` still drives separation); the Premium and Standard segments partially merge because stay duration is the only remaining continuous differentiator without ADR signal. This confirms that `is_non_refund` provides an independent structural axis, while `total_spend_log` contributes the price-level separation between Premium and Standard. The withADR representation is retained as primary because the research question explicitly includes pricing characteristics.

5.5 PCA representation study (Module E4)

PCA applied to the 7D withADR matrix before K-Means ($k = 3$):

Representation	Var. explained	Silhouette	CH	ARI vs original
Original (7D)	1.000	0.452	63,837	1.000
PCA-2	0.773	—	—	—
PCA-3	0.883	—	—	—
PCA-4	0.972	—	—	—

(Fill in PCA silhouette and ARI values from notebook output after running cell `pca4kmeans.`)

Key caveat: variance preservation does not guarantee cluster-separating directions are retained. `is_non_refund` has variance $0.12 \times 0.88 = 0.106$ — far lower than `total_spend_log` — and is therefore compressed into later principal components. If ARI vs original remains near 1.0 with lower silhouette, PCA compressed the space without improving structure, confirming the original 7D representation is preferable.

6 Discussion

Which result best answers the segmentation question? K-Means $k = 3$ on the withADR representation best answers the question. It produces three stable, interpretable segments (ARI 0.989) that align with intuitive booking archetypes: commitment-driven planners, standard flexible guests, and high-value leisure visitors. Hierarchical Ward independently recovers the same structure, strengthening confidence in the solution.

How did preprocessing choices influence results? The `is_non_refund` $\times 3.0$ weight is the single most consequential preprocessing decision: without it, the Non-Refundable segment does not emerge cleanly because the binary feature’s natural scale (0/1) is dominated by the continuous features. The scaler sensitivity analysis shows that switching from RobustScaler to StandardScaler on numerical features produces only a marginal silhouette change, confirming robustness to this choice. MinMax scaling on all features (including binary) artificially inflates scores and is not appropriate here.

Are the clusters coherent, stable, and useful? Coherence: Silhouette 0.452 is above the weak-structure threshold; intra-cluster variance of the Non-Refundable segment is the lowest (0.988), confirming it is the tightest cluster. The Premium segment has the highest intra-cluster variance (5.483), reflecting heterogeneity in stay lengths and rates within the high-spend group. Stability: near-perfect ARI across seeds. Interpretability: each segment maps to a distinct strategic action (yield management for Non-Refundable; upselling for Premium; flexibility-based conversion for Standard).

Decision-relevant recommendations.

- **Non-Refundable Advance Booker:** offer early-bird rate locks and package upgrades at booking time; cancellation risk is minimal.
- **Standard Flexible City Guest:** targeted last-minute promotional rates and parking-inclusive packages may increase conversion.
- **Premium Resort Long-Stay:** personalised pre-arrival upsells (room upgrades, spa, F&B packages) given above-average special requests.

What should not be concluded. Clusters are defined at booking level, not guest level; a returning guest may appear in different segments across stays. The absence of ground-truth labels means segment validity cannot be externally verified. The premium ADR separation between C0 and C1 may partly reflect hotel type (resort vs city) rather than guest willingness-to-pay. Finally, the Manual weights (`is_non_refund` $\times 3$, `meal` $\times 0.5$) embed subjective assumptions that should be revisited with domain expert input.

7 Limitations, Ethics, and Reproducibility

Limitations.

- No ground-truth segment labels; internal validity indices measure geometric compactness, not business relevance.
- Unit of analysis is the booking record, not the guest; one guest may appear in multiple segments across stays.
- Ordinal meal-plan encoding assumes equal spacing between plan levels; a one-hot encoding would be more conservative.
- `lead_time_rank` uses quantile binning which may lose information relative to a log-transformed continuous feature.
- The dataset covers two Portuguese hotels (2015–2017); generalisability to other markets or time periods is untested.
- iK-Means runtime was prohibitive on the full dataset in the project timeline; results should be treated as exploratory.

Reproducibility.

- Repository tag: `Submission`
- Experiment log: `experiments.csv` (schema: `date/run_id`, `representation_id`, `model`, `parameters`, `seed`, `silhouette`, `calinski`, `davies`, `diagnostics`, `notes`)
- Environment: `environment.yml`
- Entry point: `run_all.py` executes the full notebook end-to-end via `nbconvert`
- Raw data: not committed; obtain from course platform, place in `data/hotel_bookings_course_release.v1`

Ethics and governance. Dataset used in accordance with course release terms. No personal identifiers are present; country of origin is used only for post-hoc profiling. Raw data not committed to the repository.

Academic integrity. Claude Code (Anthropic) was consulted during development for methodological discussion, syntax review, and code structuring. All analytical decisions, model choices, interpretations, and written content were produced and validated by the team. The tool was not used to generate substantive analyses or report text autonomously.

8 Conclusion

This study partitioned 117,430 hotel booking records into three stable guest segments using K-Means clustering on a mixed-type pre-stay feature set. The three-cluster solution (Silhouette = 0.452, ARI = 0.989 ± 0.007 across 10 seeds, corroborated by independent Hierarchical Ward clustering) identifies a Non-Refundable Advance Booker, a Standard Flexible City Guest, and a Premium Resort Long-Stay Guest — each with distinct revenue management implications. The noADR comparison confirms that booking-commitment behaviour (`is_non_refund`) drives cluster separation independently of price. Future work should validate segments against actual revenue outcomes, test alternative meal-plan encodings, and extend the analysis to multi-hotel or multi-year datasets.

References

- [1] António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. <https://doi.org/10.1016/j.dib.2018.11.126>
- [2] Mirkin, B. (2005). *Clustering for Data Mining: A Data Recovery Approach*. Chapman & Hall/CRC.
- [3] Kaufman, L., & Rousseeuw, P. J. (1990). *Finding Groups in Data: An Introduction to Cluster Analysis*. Wiley.
- [4] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Appendix

A. Feature table (with ADR representation)

Feature	Type	Scaling	Weight
total_spend_log	Continuous	RobustScaler	$\times 1.0$
total_nights	Continuous	RobustScaler	$\times 1.0$
is_non_refund	Binary	Raw	$\times 3.0$
total_of_special_requests	Count	Raw	$\times 1.0$
required_car_parking_spaces	Binary	Raw	$\times 1.0$
meal_encoded	Ordinal (0–3)	Raw	$\times 0.5$
lead_time_rank	Ordinal (0–2)	Raw	$\times 1.0$

B. Stability analysis summary (K-Means, 10 seeds)

k	Mean ARI	Std ARI	Mean Silhouette	Std Silhouette
2	0.9975	0.0021	—	—
3	0.9894	0.0075	—	—
4	0.9942	0.0057	—	—
5	0.9803	0.0138	—	—

(Fill in mean/std silhouette values from notebook stability cell output.)

C. Scaler sensitivity (numerical features only, $k = 3$)

Scaler (numerical only)	k	Silhouette	CH	DB
RobustScaler (main pipeline)	3	—	—	—
StandardScaler	3	—	—	—

(Fill in from notebook sensitivity cell output.)